

Day 38 - DNS

Sunday, September 6, 2026

10:46 AM

DNS Info

DNS General Information

Domain Name System (DNS) is a protocol that's used to **resolve** human-readable domain names (Ex. youtube.com) to IP addresses that can be understood by devices.

- It uses port **53** for communication through either the **TCP** or **UDP** protocol.
- When you type a domain name (Ex. google.com) into a web browser, your device will ask a **DNS** server for the IP address of that domain, so that it can reach it.
- The **DNS** server(s) your device uses can be manually configured or learned via **Dynamic Host Configuration Protocol (DHCP)**.
- If the configured **DNS** server is an external server, when a device sends domain lookup requests, your router will simply forward the packets to the **DNS** server. (Therefore, no **DNS** configuration is required on the router. It is optional)

When does DNS use TCP or UDP on port **53**?

- Standard **DNS** queries/responses typically use the **UDP** protocol.
- **TCP** is used for **DNS** messages greater than **512** bytes
- Port **53** is always used in both cases.

DNS Record Types

There are multiple DNS record types which store a variety of information. The most important ones for the CCNA are:

1. **DNS 'A' Records**

- Used to map names to **IPv4** addresses.

2. **DNS 'AAAA' Records**

- Used to map names to **IPv6** addresses.

Devices will save the DNS server's responses to a locally stored DNS **cache**, meaning that they won't have to query the server every single time they want to access a particular destination.

DNS Configuration on Windows

DNS settings and other networking settings can be seen and configured through the following commands on the **Windows** operating system:

[Transclude of IP-Configuration-Commands#^ipconfig-general-windows](#)

[Transclude of IP-Configuration-Commands#^ipconfig-dns-windows](#)

- **ipconfig**
 - Provides basic information about the current device's networking configuration.
- **ipconfig /all**
 - Provides advanced information about the current device's networking configuration, including the IP address, **default gateway** and **DNS** server.
- **ipconfig /displaydns ^dns-cache**
 - Shows the locally stored **DNS** cache on the device.
- **ipconfig /flushdns**
 - Clears the current **DNS** cache on the device.
- **nslookup DOMAIN-NAME**
 - Sends a **DNS** query from the host to a **DNS** server, in order to look up the actual IP address of the specified domain.

DNS Configuration on Cisco IOS

DNS doesn't need to be configured on routers. When hosts in the network use DNS, the routers will simply forward the DNS messages like any other packet to the DNS server(s).

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However, a Cisco router can be configured as a DNS server, as well as a DNS client.

- *show hosts*
 - Shows all configured and cached **DNS** hosts as well as their IP addresses.
- *ip dns server*
 - Configures the router to act as a **DNS** server.
- *ip host HOST-NAME IP-ADDRESS*
 - Configures a list of hostname/IP address mappings.
- *ip name-server IP-ADDRESS*
 - Configures a **DNS** server that the router will query if the requested record isn't available in its *host* table.
- *ip domain lookup* or *ip domain-lookup*
 - Enables the router to perform **DNS** queries to the configured **DNS** server.
 - (Enabled by default)
- *ip domain name DOMAIN-NAME* or *ip domain-name DOMAIN-NAME*
 - Configures the default *domain* name.
 - This will be automatically appended to any hostnames without a specified *domain*.
 - (Ex. if the *domain* name is *kruber.party*, then *PC1* would become reachable at *PC1.kruber.party*)